

Assignment_GroupBy_CountProduct

Summary

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Tryout > 1: Question 1

Problem Statement

Display the total number of products in each review category.

For the given sample data, the following is the expected output.

Sample Output:

REVIEW	NUMBER_OF_PRODUCTS
Excellent	3
Good	3
Bad	2

Database Structure and Data:

Table Structure: Vendor (This table provides the details of all the vendors)

Column Name	Data Type and Size	Constraint	Description
VendorId	VARCHAR(5)	PRIMARY KEY, CHECK	unique Id for each vendor, should start with 'V'

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Database Structure and Data:

Table Structure: Vendor (This table provides the details of all the vendors)

Column Name	Data Type and Size	Constraint	Description
VendorId	VARCHAR(5)	PRIMARY KEY, CHECK	unique Id for each vendor, should start with 'V'
VendorName	VARCHAR(30)	NOT NULL	name of the vendor
VendorContactNo	NUMERIC(10)	UNIQUE	contact number of the vendor
Rating	INTEGER	CHECK	rating of vendor, must be in the range of 0 to 5 (both inclusive)

Table Structure: Product (This table provides the details of all the products)

Column Name	Data Type and Size	Constraint	Description

Tryout > 1: Question 1

Table Structure: Product (This table provides the details of all the products)

Column Name	Data Type and Size	Constraint	Description
ProductId	VARCHAR(5)	PRIMARY KEY, CHECK	unique Id for every product, should start with 'P'
PName	VARCHAR(30)	NOT NULL	name of the product
Brand	VARCHAR(20)	NOT NULL	brand of the product
PCategory	VARCHAR(20)		category of the product
Color	VARCHAR(10)		color of the product
Review	VARCHAR(10)	CHECK	review of the product, should be 'Bad' or 'Good' or 'Excellent'

Table Structure: Customer (This table provides the details of all the customers)

Column Name	Data Type and Size	Constraint	Description

Tryout > 1: Question 1

Table Structure: Customer (This table provides the details of all the customers)

Column Name	Data Type and Size	Constraint	Description
CustId	VARCHAR(5)	PRIMARY KEY, CHECK	unique Id of the customer, must begin with character 'C'
CustName	VARCHAR(10)	NOT NULL	name of the customer
Age	NUMERIC(2)	CHECK	age of the customer, should be between 0 and 99 (both inclusive)
CustLocation	VARCHAR(20)		location, in which the customer resides
Email	VARCHAR(30)	UNIQUE	email id of the customer
Gender	CHAR(1)	CHECK	gender of the customer, should be 'M' or 'F'

Table Structure: VendorOffering (This table provides the details of all the products)

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Table Structure: VendorOffering (This table provides the details of all the products offered by different vendors)

offered by different vendors)				
Column Name	Data Type and Size	Constraint		Description
VendorId	VARCHAR(5)	FOREIGN KEY	PRIMARY KEY	existing VendorId present in Vendor Table
ProductId	VARCHAR(5)	FOREIGN KEY, NOT NULL		existing ProductId present in Product Table
Price	NUMERIC(7,2)	NOT NULL, CHECK		price offered for the product by the vendor in INR, should be greater than zero

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Table Structure: Invoice (This table provides the details of all the products purchased by the customers)

Column Name	Data Type and Size	Constraint	Description
InvoiceId	VARCHAR(5)	PRIMARY KEY, CHECK	unique InvoiceId for every Invoice, should start with 'IN'
CustId	VARCHAR(5)	FOREIGN KEY	existing CustId present in Customer table
VendorId	VARCHAR(5)	FOREIGN KEY	existing {VendorId,ProductId} present in VendorOffering table
ProductId	VARCHAR(5)		
QuantityPurchased	NUMERIC(2)	NOT NULL	quantity (number of products) purchased by the customer
Discount	NUMERIC(3,1)	NOT NULL	discount offered on the purchase (in percentage)

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			percentage)
PurchaseDate	DATE	NOT NULL	date of purchase

Table Data: Vendor

VENDORID	VENDORNAME	VENDORCONTACTNO	RATING
V101	Wendy Creations	8976567890	4
V102	CrazyNuts	NULL	NULL
V103	FlyingHigh Retailers	8126567890	5
V104	Global Traders	8005432556	4
V105	Caretech Enterpises	8116567890	3
V106	Infinity Solutions	8056781234	NULL
V107	BrightWorks Technology	8045678902	3
V108	Walgreens Retailers	8034567891	4

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v108	waigreens Retailers	8034567891	4
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Table Data: Customer

CUSTID	CUSTNAME	AGE	CUSTLOCATION	EMAIL	GENDER
C101	Nancy	23	Plattsburg	nancy@gmail.com	F
C102	Bob	33	Ithaca	bob@gmail.com	M
C103	Clara	NULL	Yonkers	clara@gmail.com	F
C104	Vincent	34	Ithaca	vincent@gmail.com	M
C105	Alex	27	Cortland	NULL	M
C106	Peter	77	Yonkers	peter@gmail.com	M
C107	Rose	54	Ithaca	rose@gmail.com	F
C108	Elina	18	Auburn	elina@gmail.com	F

Table Data: Product

PRODUCTID	PNAME	BRAND	CATEGORY	COLOR	RETAIL
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Table Data: Product

PRODUCTID	PNAME	BRAND	PCATEGORY	COLOR	REVIEW
P101	High Boots	Diesel	Shoes	Blue	Good
P102	Mens Formal Shirt	Terrain	Clothing	Turquoise	Bad
P103	Pearl Earring	Essence	Accessories	Yellow	Excellent
P104	Kids Crocks	Sparx	NULL	Pink	Excellent
P105	Kids Shirt	Cosmos	Clothing	White	Bad
P106	Sporty Sneakers	Reebok	Shoes	White	Good
P107	Womens Jacket	NorthFace	Clothing	NULL	Good

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P103	Pearl Earring	Essence	Accessories	Yellow	Excellent
P104	Kids Crocks	Sparx	NULL	Pink	Excellent
P105	Kids Shirt	Cosmos	Clothing	White	Bad
P106	Sporty Sneakers	Reebok	Shoes	White	Good
P107	Womens Jacket	NorthFace	Clothing	NULL	Good
P108	Leather Wallet	Luxurio	Accessories	Brown	Excellent

Table Data: VendorOffering

VENDORID	PRODUCTID	PRICE
V101	P101	3000.40

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Table Data: VendorOffering

VENDORID	PRODUCTID	PRICE
V101	P101	3000.40
V102	P102	1200.30
V102	P104	1000.80
V103	P101	2800.10
V103	P103	400.20
V104	P104	900.10
V104	P105	500.80
V105	P105	445.00
V107	P107	90.50
V108	P106	699.99

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Table Data: Invoice

INVOICEID	CUSTID	VENDORID	PRODUCTID	QUANTITYPURCHASED	DISCOUNT
IN101	C101	V101	P101	2	12.5
IN102	C102	V102	P102	3	13.0
IN103	C103	V103	P103	1	15.2
IN104	C104	V104	P104	1	10.0
IN105	C103	V105	P105	2	5.0
IN106	C106	V102	P104	7	10.8
IN107	C108	V103	P101	4	2.0
IN108	C107	V104	P105	2	3.3
IN109	C108	V108	P106	8	7.0
IN110	C101	V103	P101	10	20.4

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ID	VENDORID	PRODUCTID	QUANTITYPURCHASED	DISCOUNT	PURCHASEDATE
1	V101	P101	2	12.5	2023-08-11
2	V102	P102	3	13.0	2023-09-23
3	V103	P103	1	15.2	2023-06-10
4	V104	P104	1	10.0	2023-09-01
3	V105	P105	2	5.0	2023-09-08
6	V102	P104	7	10.8	2023-10-02
8	V103	P101	4	2.0	2023-10-12
7	V104	P105	2	3.3	2023-06-12
8	V108	P106	8	7.0	2023-07-18
1	V103	P101	10	20.4	2023-08-18

Assignment_GroupBy_MonthWiseCount

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Tryout > 1: Question 1

Problem Statement

Find the number of subscribers that have subscribed in each month. Display the month as "MONTH" (column alias) and number of subscribers as "SUBSCRIBERCOUNT" (column alias).

The following records will appear as a part of output along with other record(s).

Sample Output:

MONTH	SUBSCRIBERCOUNT
November	3
September	1

Database Structure and Data:

Table Structure: Publisher (This table provides the details of all the publishers)

Column Name	Data Type and Size	Constraint	Description
PublisherId	VARCHAR(4)	PRIMARY KEY,	unique id of the publisher, must begin with character 'P' For

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Tryout > 1: Question 1

Table Structure: Publisher (This table provides the details of all the publishers)

Column Name	Data Type and Size	Constraint	Description
PublisherId	VARCHAR(4)	PRIMARY KEY, CHECK	unique id of the publisher, must begin with character 'P'. For ex. P101, P102
PublisherName	VARCHAR(20)	NOT NULL, UNIQUE	name of the publisher, cannot be more than 20 characters
Rating	NUMERIC(3,1)	NOT NULL, CHECK	rating given to the publisher, should be in the range of 1 to 5 (both inclusive)
ContactNo	NUMERIC		contact number of the publisher

Table Structure: Journal (This table provides the details of all the journals)

Column Name	Data Type and Size	Constraint	Description
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Table Structure: Journal (This table provides the details of all the journals)

Column Name	Data Type and Size	Constraint	Description
JournalId	VARCHAR(4)	PRIMARY KEY, CHECK	unique id of the journal, must begin with character 'J'. For ex. J201, J202
JournalName	VARCHAR(25)	NOT NULL, CHECK	name of the journal, cannot be more than 25 characters
Genre	VARCHAR(20)	NOT NULL, CHECK	genre of the journal, cannot be more than 20 characters, and should be either

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Genre	VARCHAR(20)	NOT NULL, CHECK	20 characters, and should be either 'Academics', 'Scientific', 'Politics' or 'Trade'
MonthlyCharges	INTEGER	NOT NULL, CHECK	monthly subscription charges for the journal (in INR), should be more than 0
PublisherId	VARCHAR(4)	FOREIGN KEY	publisher of the journal, refers to existing PublisherId in the publisher table

Table Structure: Customer (This table provides the details of all the customers)

Column Name	Data Type and Size	Constraint	Description
CustomerId	VARCHAR(4)	PRIMARY KEY, CHECK	unique id of the customer, must begin with character 'C'. For ex. C101, C102
CustomerName	VARCHAR(15)	NOT NULL	name of the customer
CustomerType	VARCHAR(10)	CHECK	type of the customer, should be either 'Regular', 'Privileged' or 'Prime'
State	VARCHAR(15)		state in which the customer resides
City	VARCHAR(15)		city in which the customer resides

Table Structure: Subscription (This table provides the details of all the subscriptions)

Column Name	Data Type and Size	Constraint	Description
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Table Structure: Subscription (This table provides the details of all the subscriptions)

Column Name	Data Type and Size	Constraint	Description
SubscriptionId	VARCHAR(4)	PRIMARY KEY, CHECK	unique id of the subscription, must begin with character 'S'. For ex. S301, S302
JournalId	VARCHAR(4)	FOREIGN KEY	existing JournalId in Journal table
CustomerId	VARCHAR(4)	FOREIGN KEY	existing CustomerId in Customer table
StartDate	DATE		starting date of subscription, default value will be the current system date
DurationInMonths	INTEGER	NOT NULL,	duration of the subscription (in months), should be

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StartDate	DATE		default value will be the current system date
DurationInMonths	INTEGER	NOT NULL, CHECK	duration of the subscription (in months), should be more than 0, default value is 3
DiscountPercent	NUMERIC		discount given to the customer on subscription (in percentage)

Table Data: Publisher

PUBLISHERID	PUBLISHERNAME	RATING	CONTACTNO
P101	Taylor & Francis	4.0	8765678932
P102	Wiley	2.0	NULL
P103	Sage Publications	2.0	7885678932

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			percentage,
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Table Data: Publisher

PUBLISHERID	PUBLISHERNAME	RATING	CONTACTNO
P101	Taylor & Francis	4.0	8765678932
P102	Wiley	2.0	NULL
P103	Sage Publications	2.0	7885678932
P104	World Scientific	3.0	9765678932
P105	Routledge	5.0	NULL
P106	Springer Nature	4.0	7685655932

Table Data: Journal

JOURNALID	JOURNALNAME	GENRE	MONTHLYCHARGES	PUBLISHERID
J201	The Hollywood Reporter	Trade	130	P105

Tryout > 1: Question 1**Table Data: Journal**

JOURNALID	JOURNALNAME	GENRE	MONTHLYCHARGES	PUBLISHERID
J201	The Hollywood Reporter	Trade	130	P105
J202	Kirkus Reviews	Trade	90	P102
J203	The Lancet	Scientific	75	P106
J204	eLife	Academics	85	P102
J205	Pulp and Paper	Trade	125	P102
J206	Acta Politica	Politics	110	P105
J207	Global Policy	Politics	105	P103
J208	LOS Biology	Academics	95	P101

Table Data: Customer

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Table Data: Customer

CUSTOMERID	CUSTOMERNAME	CUSTOMERTYPE	STATE	CITY
C101	Sam	Regular	New York	Manhattan
C102	Jimmy	Privileged	Illinois	Chicago
C103	Michael	Prime	Texas	Dallas
C104	Oliver	Regular	Illinois	Chicago
C105	Clara	Regular	Arizona	Phoenix
C106	Amelia	Privileged	Arizona	Cottonwood
C107	Robert	Regular	Illinois	Galesburg

Table Data: Subscription

SUBSCRIPTIONID	JOURNALID	CUSTOMERID	STARTDATE	DURATIONINMONTHS
			2023-11-	

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Table Data: Subscription

SUBSCRIPTIONID	JOURNALID	CUSTOMERID	STARTDATE	DURATIONINMONTHS	DISCOUNT
S301	J202	C107	2023-11-22	8	0%
S302	J205	C105	2023-09-10	6	0%
S303	J201	C102	2023-10-15	4	0%
S304	J207	C103	2023-02-25	12	0%
S305	J203	C101	2023-11-22	12	0%
S306	J206	C105	2023-11-22	10	0%
S307	J202	C106	2023-01-12	6	0%
S308	J204	C107	2023-06-29	8	0%
S309	J205	C103	2023-01-12	5	0%

Tryout > 1: Question 1

3: Subscription

SubscriptionID	JournalID	CustomerID	StartDate	DurationInMonths	DiscountPercent
101	J202	C107	2023-11-22	8	10
102	J205	C105	2023-09-10	6	7
103	J201	C102	2023-10-15	4	0
104	J207	C103	2023-02-25	12	15
105	J203	C101	2023-11-22	12	15
106	J206	C105	2023-11-22	10	10
107	J202	C106	2023-01-12	6	7
108	J204	C107	2023-06-29	8	10
109	J205	C103	2023-01-12	5	0

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S302	J205	C105	2023-09-10	6	
S303	J201	C102	2023-10-15	4	
S304	J207	C103	2023-02-25	12	
S305	J203	C101	2023-11-22	12	
S306	J206	C105	2023-11-22	10	
S307	J202	C106	2023-01-12	6	
S308	J204	C107	2023-06-29	8	
S309	J205	C103	2023-01-12	5	
S310	J207	C102	2023-08-17	2	

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Q02	J205	C105	2023-09-10	6	7
Q03	J201	C102	2023-10-15	4	0
Q04	J207	C103	2023-02-25	12	15
Q05	J203	C101	2023-11-22	12	15
Q06	J206	C105	2023-11-22	10	10
Q07	J202	C106	2023-01-12	6	7
Q08	J204	C107	2023-06-29	8	10
Q09	J205	C103	2023-01-12	5	0
Q10	J207	C102	2023-08-17	2	0

Assignment_GroupBy_TwoColumns

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Tryout > 1: Question 1

Problem Statement

Retrieve the data of those subscription(s) with duration more than 3 months and the customer has got same discount percent again on another subscription. Display the Customerid, Discountpercent and average DurationInMonths of those subscriptions.

For the given sample data, the following is the expected output.

Expected Output:

CUSTOMERID	DISCOUNTPERCENT	AVG_DURATION
C107	10	8.00000

Database Structure and Data:

Table Structure: Publisher (This table provides the details of all the publishers)

Column Name	Data Type and Size	Constraint	Description
PublisherId	VARCHAR(4)	PRIMARY KEY, CHECK	unique id of the publisher, must begin with character 'P'. For ex. P101, P102
PublisherName	VARCHAR(20)	NOT NULL, UNIQUE	name of the publisher, cannot be more than 20 characters
Rating	NUMERIC(2,1)	NOT NULL,	rating given to the publisher, should be in the range of 1 to

Assignment_GroupBy_Having

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Tryout > 1: Question 1

Problem Statement

Display the name of product category(ies) for which more than one product exists.

For the given sample data, the following record will feature as part of the output along with other record(s).

Sample Output:

PCATEGORY
Clothing

Database Structure and Data:

Table Structure: Vendor (This table provides the details of all the vendors)

Column Name	Data Type and Size	Constraint	Description
VendorId	VARCHAR(5)	PRIMARY KEY, CHECK	unique Id for each vendor, should start with 'V'
VendorName	VARCHAR(30)	NOT NULL	name of the vendor
VendorContactNo	NUMERIC(10)	UNIQUE	contact number of the vendor
Rating	INTEGER	CHECK	rating of vendor, must be in the range of 0 to 5 (both

Assignment_GroupBy_GenreWiseAvgMonthlyCharges

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Tryout > 1: Question 1

Problem Statement

For each Genre whose average monthly charge is more than 100, display the Genre and its corresponding average monthly charges as "AVG_CHARGES" (column alias).

For the given sample data, the following is the expected output.

Sample Output:

GENRE	AVG_CHARGES
Trade	115.00000
Politics	107.50000

Database Structure and Data:

Table Structure: Publisher (This table provides the details of all the publishers)

Column Name	Data Type and Size	Constraint	Description
PublisherId	VARCHAR(4)	PRIMARY KEY, CHECK	unique id of the publisher, must begin with character 'P'. For ex. P101, P102
PublisherName	VARCHAR(20)	NOT NULL, UNIQUE	name of the publisher, cannot be more than 20 characters
			rating given to the publisher

Assignment_GroupBy_FilterMonthWiseCount

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Tryout > 1: Question 1

Problem Statement

Identify the month(s) in which more than one subscription of duration 6 or more (months) has been taken. Display abbreviated name of month as "MONTH" (column alias) and total number of such subscriptions taken in that month as "TOTAL_SUBSCRIPTIONS" (column alias)

For the given sample data, the following is the expected output.

Expected Output:

MONTH	TOTAL_SUBSCRIPTIONS
Nov	3

Database Structure and Data:

Table Structure: Publisher (This table provides the details of all the publishers)

Column Name	Data Type and Size	Constraint	Description
PublisherId	VARCHAR(4)	PRIMARY KEY, CHECK	unique id of the publisher, must begin with character 'P'. For ex. P101, P102
PublisherName	VARCHAR(20)	NOT NULL, UNIQUE	name of the publisher, cannot be more than 20 characters
Rating	NUMERIC(2, 1)	NOT NULL,	rating given to the publisher, should be in the range of 1 to

Assignment_GroupBy_FilterAvgDiscount

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Tryout > 1: Question 1

Problem Statement

Retrieve the JournalId and average DiscountPercent of the journals purchased before '2023-11-01' as "AVG_DIS" (column alias).

For the given sample data, the following records will feature as part of the output along with other record(s).

Sample Output:

JOURNALID	AVG_DIS
J205	3.50000
J202	7.00000

Database Structure and Data:

Table Structure: Publisher (This table provides the details of all the publishers)

Column Name	Data Type and Size	Constraint	Description
PublisherId	VARCHAR(4)	PRIMARY KEY, CHECK	unique id of the publisher, must begin with character 'P'. For ex. P101, P102
PublisherName	VARCHAR(20)	NOT NULL, UNIQUE	name of the publisher, cannot be more than 20 characters
			rating given to the publisher

Assignment_GroupBy_AvgMonthlyCharges

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Tryout > 1: Question 1

Problem Statement

For every Genre except 'Trade', display the Genre and average monthly charges of the journals of that Genre as "AVERAGE CHARGE" (column alias), only if, the identified average charge(s) is less than 100.

For the given sample data, the following is the expected output.

Sample Output:

GENRE	Average Charge
Academics	90.00000
Scientific	75.00000

Database Structure and Data:

Table Structure: Publisher (This table provides the details of all the publishers)

Column Name	Data Type and Size	Constraint	Description
PublisherId	VARCHAR(4)	PRIMARY KEY, CHECK	unique id of the publisher, must begin with character 'P'. For ex. P101, P102
PublisherName	VARCHAR(20)	NOT NULL, UNIQUE	name of the publisher, cannot be more than 20 characters

Assignment_GroupBy_JournalWiseSubscription

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Tryout > 1: Question 1

Problem Statement

For each journal, identify the total number of subscriptions for that journal. Display JournalId and total number of subscriptions as "TOTALSUBSCRIPTIONS" (column alias).

For the given sample data, the following records will feature as part of the output along with other record(s).

Sample Output:

JOURNALID	TOTALSUBSCRIPTIONS
J203	1
J205	2

Database Structure and Data:

Table Structure: Publisher (This table provides the details of all the publishers)

Column Name	Data Type and Size	Constraint	Description
PublisherId	VARCHAR(4)	PRIMARY KEY, CHECK	unique id of the publisher, must begin with character 'P'. For ex. P101, P102
PublisherName	VARCHAR(20)	NOT NULL, UNIQUE	name of the publisher, cannot be more than 20 characters

Assignment_GroupBy_FilterAvgCharges

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Tryout > 1: Question 1

Problem Statement

Identify the publisher(s) whose average monthly charges of journals having Genre as 'Trade' or 'Politics' is less than 110. Display PublisherId and average monthly charges of the publisher as "MONTHLY_CHARGES" (column alias).

For the given sample data, the following is the expected output.

Sample Output:

PUBLISHERID	MONTHLY_CHARGES
P102	107.50000
P103	105.00000

Database Structure and Data:

Table Structure: Publisher (This table provides the details of all the publishers)

Column Name	Data Type and Size	Constraint	Description
PublisherId	VARCHAR(4)	PRIMARY KEY, CHECK	unique id of the publisher, must begin with character 'P'. For ex. P101, P102
PublisherName	VARCHAR(20)	NOT NULL, UNIQUE	name of the publisher, cannot be more than 20 characters

Assignment_GroupBy_StartDate

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Tryout > 1: Question 1

Problem Statement

Identify the dates on which average discount percent offered on journals that have been subscribed for atleast 6 months is more than 10. Display the identified subscription StartDate(s) and the average discount percent offered on that date as "AVG_DISCOUNT" (column alias).

For the given sample data, the following is the expected output.

Sample Output:

STARTDATE	AVG_DISCOUNT
2023-02-25	15.00000
2023-11-22	11.66666

Database Structure and Data:

Table Structure: Publisher (This table provides the details of all the publishers)

Column Name	Data Type and Size	Constraint	Description
PublisherId	VARCHAR(4)	PRIMARY KEY, CHECK	unique id of the publisher, must begin with character 'P'. For ex. P101, P102
PublisherName	VARCHAR(20)	NOT NULL, UNIQUE	name of the publisher, cannot